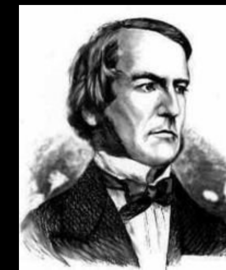
Introduction to Artificial Intelligence

Some slides are based on slides by Eakta Jain

AI has a rich history

- Philosophy
- Mathematics
- Economics
- Neuroscience
- Psychology
- Control Theory
- John McCarthy, coined the term in 1956



George Boole

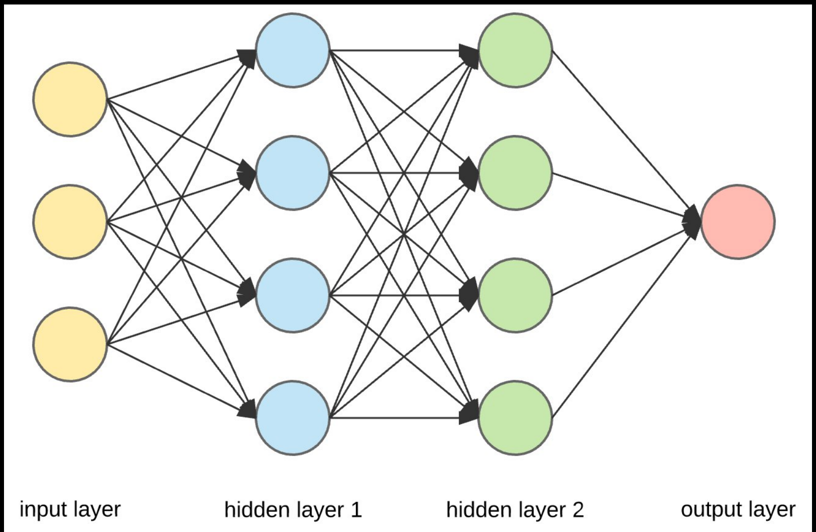
Gerolamo Cardano

Norbert Wiener

John McCarthy

What is artificial intelligence?


CHESS CHAMPION 2150L
Radio Shack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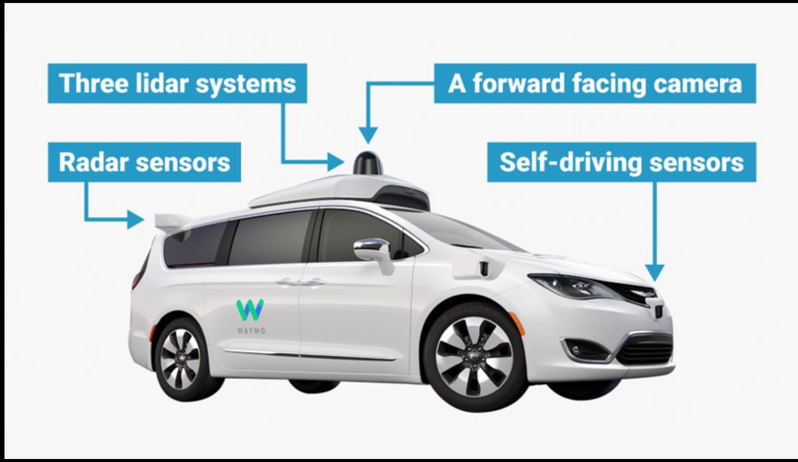
Three lidar systems
A forward facing camera
Radar sensors
Self-driving sensors


HBP
The Human Brain Project

What is Artificial Intelligence?

- Machines that think?
  - What does it mean *to think*?

- Machines that act intelligently?

  - What does it mean to *act intelligently*?

- Why “artificial”? Is there any essential difference between the intelligence displayed by biological entities and machines?

There is no common or generally recognized definition of the term *artificial intelligence*

Dictionary

Enter a word, e.g. 'pie'

artificial intelligence

noun

the theory and development of computer systems able to perform tasks normally requiring human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages.

Translations, word origin, and more definitions

Our human intuition is not a good yardstick

Which problem is harder?

What is Artificial Intelligence?

	HUMAN	RATIONAL
THOUGHT	Systems that think like humans	Systems that think rationally
BEHAVIOUR	Systems that act like humans	Systems that act rationally

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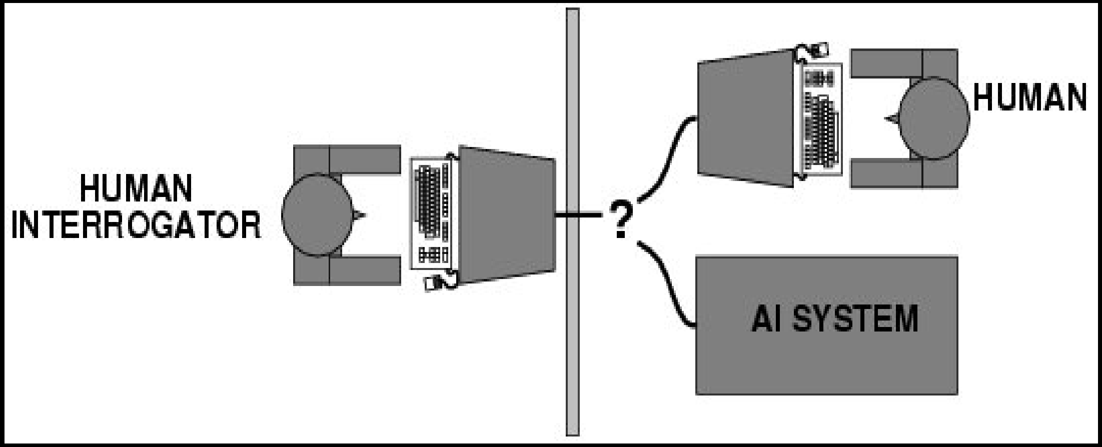
Systems that act like humans

- Human beings are intelligent
- To be called intelligent, a machine must produce responses that are indistinguishable from those of a human.

Alan Turing

The Turing Test



Turing (1950), “Computing Machinery and Intelligence”

Systems that act like humans

These cognitive tasks include:

- *Natural language processing*
  - for communication with human
- *Knowledge representation*
  - to store information effectively and efficiently
- *Automated reasoning*
  - to retrieve and answer questions using the stored information
- *Machine learning*
  - to adapt to new circumstances
- *Common sense*
  - have a good understanding of how the world works from a human perspective

The “Total Turing Test”

- Includes two more issues:
  - *Computer vision*
    - to perceive objects (seeing)
  - *Actuation*
    - to manipulate objects (acting)

What is Artificial Intelligence ?

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Systems that think like humans: cognitive modeling

- Humans as observed from ‘inside’

- How do we know how humans think?
  - Introspection vs. psychological experiments

- Cognitive science

- “The exciting new effort to make computers think … machines with *minds* in the full and literal sense” (Haugeland)

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Systems that think ‘rationally’

- “Laws of thought“ (beginning with Plato and Aristotle) - if premises are correct, conclusion will also be. Example:

  “Socrates is a man; all men are mortal; therefore, Socrates is mortal”
- Logic cannot express everything (e.g., uncertainty)
  - Statistical inference, game theory, economics

- A perfect rational approach is often not feasible in terms of computational resources
  - Bounded rationality: constraints (time, computational resources, etc.) are considered.

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Important term: “rational agent”

- Rationality is an ideal concept of intelligence:

“*A system is rational if it does the right thing*.”
- Russell and Norvig

- A <u>rational agent</u> is one that takes the optimal actions, based on its knowledge, to achieve its goal.

Example. Assume I don't like to get wet (my goal), so I bring an umbrella (my action). Do I behave rationally?

- The answer is dependent on my knowledge and belief
- If I've heard the forecast for rain and I believe it, then bringing the umbrella is rational.
- If I've not heard the forecast for rain and I do not believe that it is going to rain, then bringing the umbrella is not rational.

Note

"Behave rationally" does not always achieve the goals successfully

Example:

- My goals – (1) not to get wet if it rains; (2) not to look stupid (such as bringing an umbrella when no rain)
- My knowledge/belief – weather forecast for rain and I believe it
- My rational behaviour – bring an umbrella
- The outcome of my behavior: If rain, then my rational behavior achieves both goals; If not rain, then my rational behaviour fails to achieve the 2nd goal

The success of my rational behavior depends on my knowledge and belief, and on the stochasticity of the world.

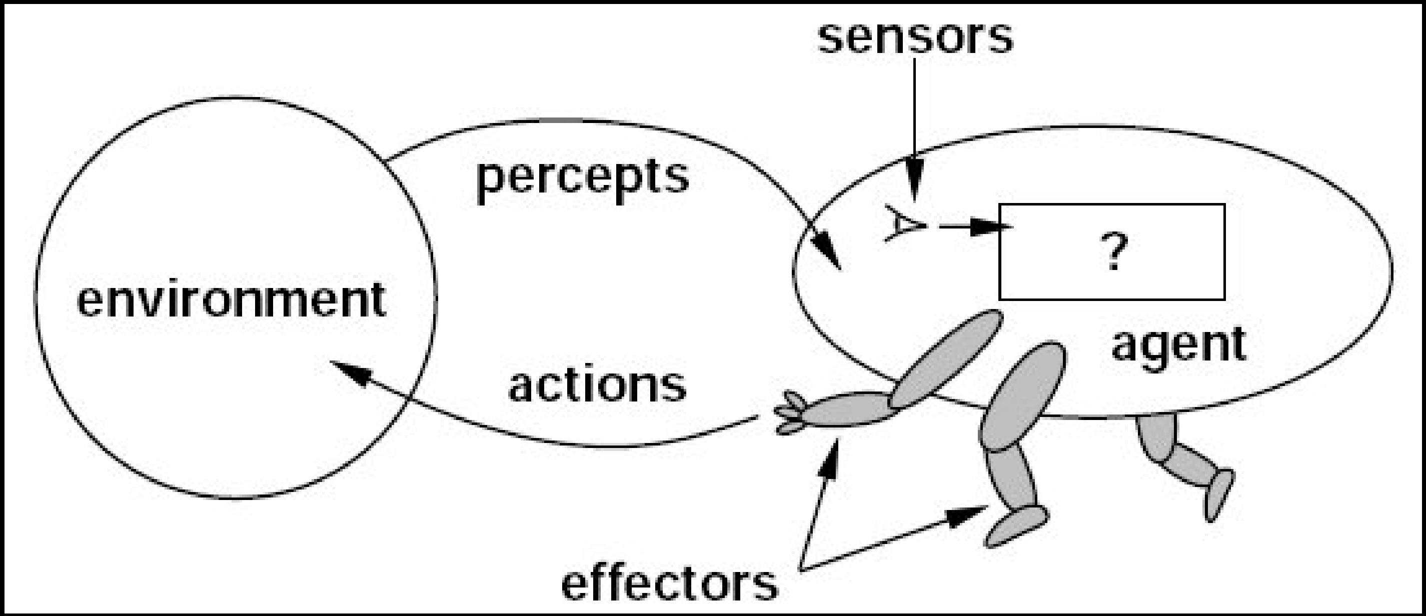
2.2.2 Omniscience, learning, and autonomy

We need to be careful to distinguish between rationality and **omniscience**. An omniscient agent knows the *actual* outcome of its actions and can act accordingly; but omniscience is impossible in reality. Consider the following example: I am walking along the Champs Elysées one day and I see an old friend across the street. There is no traffic nearby and I'm not otherwise engaged, so, being rational, I start to cross the street. Meanwhile, at 33,000 feet, a cargo door falls off a passing airliner,[2] and before I make it to the other side of the street I am flattened. Was I irrational to cross the street? It is unlikely that my obituary would read "Idiot attempts to cross street."

Artificial intelligence: A Modern Approach p. 38

An Agent

- ‘Anything’ that can gather information about its environment and take action based on that information.



Sensors, effectors and environment here?

Rational agents

- An agent is an entity that perceives and acts
- Abstractly, an agent is a function, *f*, from percept histories, P*, to actions, A:

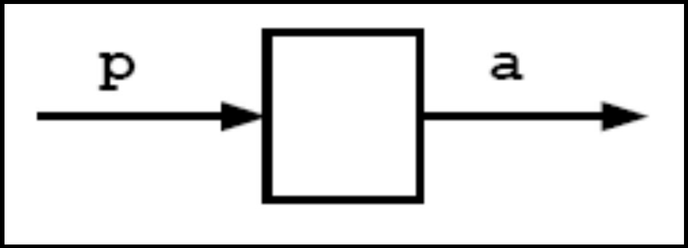
$$f: P^* \rightarrow A$$

- For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance.

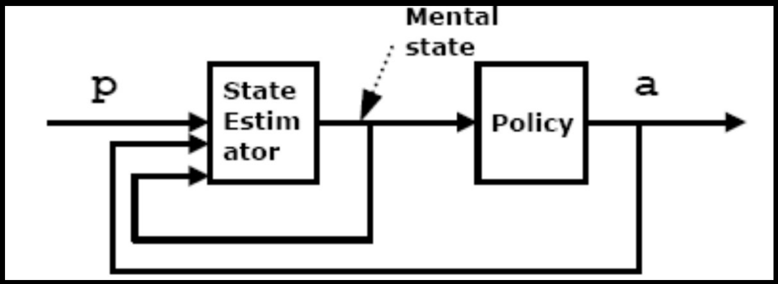
- Caveat: computational limitations make perfect rationality unachievable
    - design best agent given the available computational resources (bounded rationality)

Reactive agents

- Reactive agents
  - No memory



- Deliberative agent
  - Agents with memory



Environments

- Accessible/inaccessible
    - Does an agent have access to the complete state of the environment?
- Deterministic/non-deterministic (stochastic)
    - Will the same action always produce the same result?
- Episodic/sequential (non-episodic)
    - Is the agent’s performance the result of a series of independent, one-shot actions (episodic environment) or does an action has consequences for future (sequential)
- Static/dynamic
    - Does the environment change independent of the agent’s actions?
- Discrete/continuous
    - Is the state space of the environment discrete or continuous?

What type of environment do autonomous mobile robots typically find themselves in?

Recap

- There is no universally agreed definition of AI.
  - "AI is whatever hasn't been done yet." (Hofstader) – results of the "The AI Effect"
- There are different objectives with AI research:

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- The concepts of … l agents is important in much AI resear…

The main topics in AI

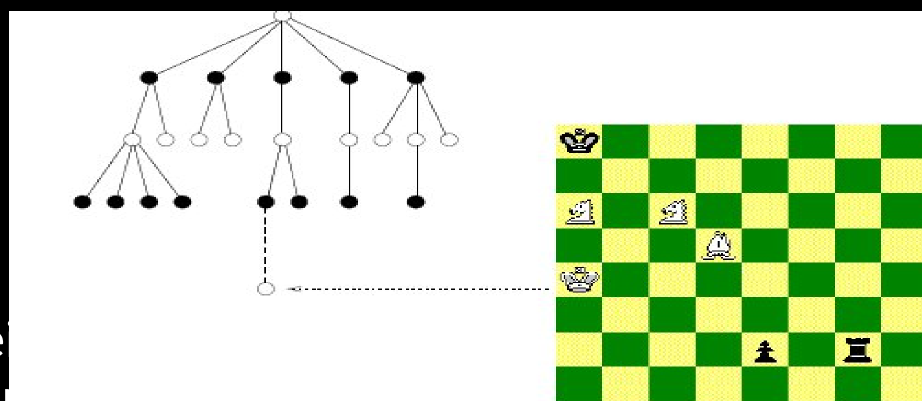
Artificial intelligence can be considered under a number of headings:

- Search (includes game playing).
- Knowledge representation.
- Planning.
- Learning.
- Natural language processing.
- Expert systems.
- Robotics.

Search

- *Search* is the fundamental technique of AI.
    - Possible answers, decisions, or courses of action are structured into an abstract space, which we then search.

- One of the early research in AI is search problem such as for game playing. Game playing can be usefully viewed as a search problem in a space defined by a fixed set of rules:



- Nodes are ei ct the adversary’s tu

- The tree of possible moves can be searched for favorable positions.

Knowledge Representation & Reasoning

- If we are going to act rationally in our environment, then we must have some way of describing that environment and drawing inferences from that representation.
  - how do we describe what we know about the world ?
  - how do we describe it *concisely* ?
  - how do we describe it so that we can get hold of the right piece of knowledge when we need it ?
  - how do we generate new pieces of knowledge ?
  - how do we deal with *uncertain* knowledge ?

Knowledge

Declarative Procedural

- Declarative knowledge deals with factoid questions (e.g. "*what is the capital of India?*")
- Procedural knowledge deals with "How" (e.g. ride a bike)

Planning

Given a set of goals, construct a sequence of actions that achieves those goals:

- often very large search space
- but most parts of the world are independent of most other parts
- often start with goals and connect them to actions
- no necessary connection between order of planning and order of execution

Learning

- Learning from existing data?
- How do we generate new facts from old?
- How do we generate new concepts?
- How do we learn to distinguish different situations in new environments?

Learning

- Incomplete information about the environment
- Changing environments
- How do we decide when to experiment to learn something new and when should we rely on existing knowledge (exploration vs. exploitation)?
- Hard for us to articulate the knowledge needed to build AI systems – e.g. try writing a program to recognize visual input like various types of flowers

Classifying learning problems

- **Supervised learning** - Given a set of input/output pairs, learn to predict the output if faced with a new input.
- **Unsupervised Learning** - Learning patterns in the input when no specific output values are supplied.
- **Reinforcement Learning** - Learn to interact with the world from the reinforcement you get.

In this course

- Fuzzy logic and control
- Reinforcement learning
- Testing AI-based algorithms and control